

Ques: Find out whether there is any setup violation or hold violation.

NOTE :- Worst case analysis

Setup time $\Rightarrow$ max delay along data path

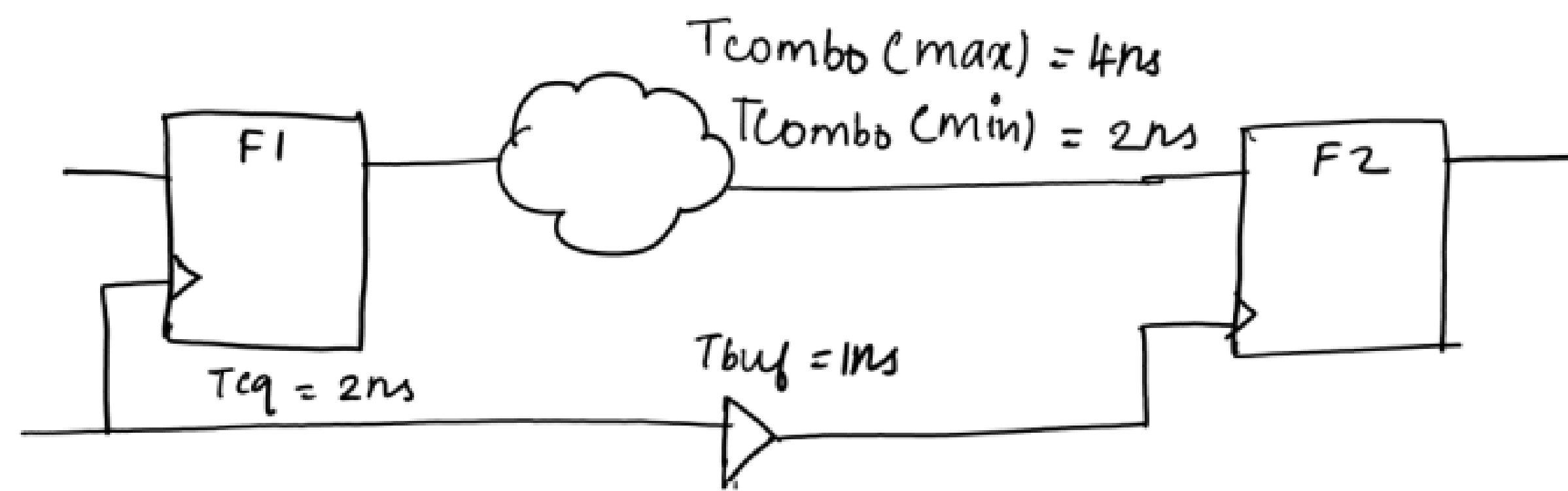
min delay along clock path

$$\text{Setup Slack} = RT_{\min} - AT_{\max}$$

$$\text{Hold slack} = AT_{\min} - RT_{\max}$$

Hold time $\Rightarrow$ min delay along data path

max delay along clock path.



$$T_{\text{setup}} = 1\text{ns}$$

$$T_{\text{hold}} = 2\text{ns}$$

$$T_{\text{clk}} = 10\text{ns}$$

Ans:- Setup time :-

$$\overbrace{\Delta_1 + T_{cq} + T_{\text{comb0}}}^{A.T} \leq \overbrace{T_{\text{clk}} - T_{\text{setup}} + \Delta_2}^{R.T}$$

$$0 + 2 + 4 \leq 10 - 1 + 1$$

$$6 \leq 10$$

$\hookrightarrow$ eqn is satisfied

$\therefore$ no setup violation.

$$\begin{aligned} \text{Setup Slack} &= RT - AT \\ &= 10 - 6 \\ &= +4\text{ns} \end{aligned}$$

$\therefore$ Setup slack is +ve, the path does not have setup violation

Hold time :- A.T

$$\overbrace{\Delta_1 + T_{cq} + T_{combo}}^{A.T} \geq \overbrace{T_{hold} + \Delta_2}^{R.T}$$
$$0 + 2 + 2 \geq 2 + 1$$

$$4 \geq 3$$

↳ eqn is satisfied

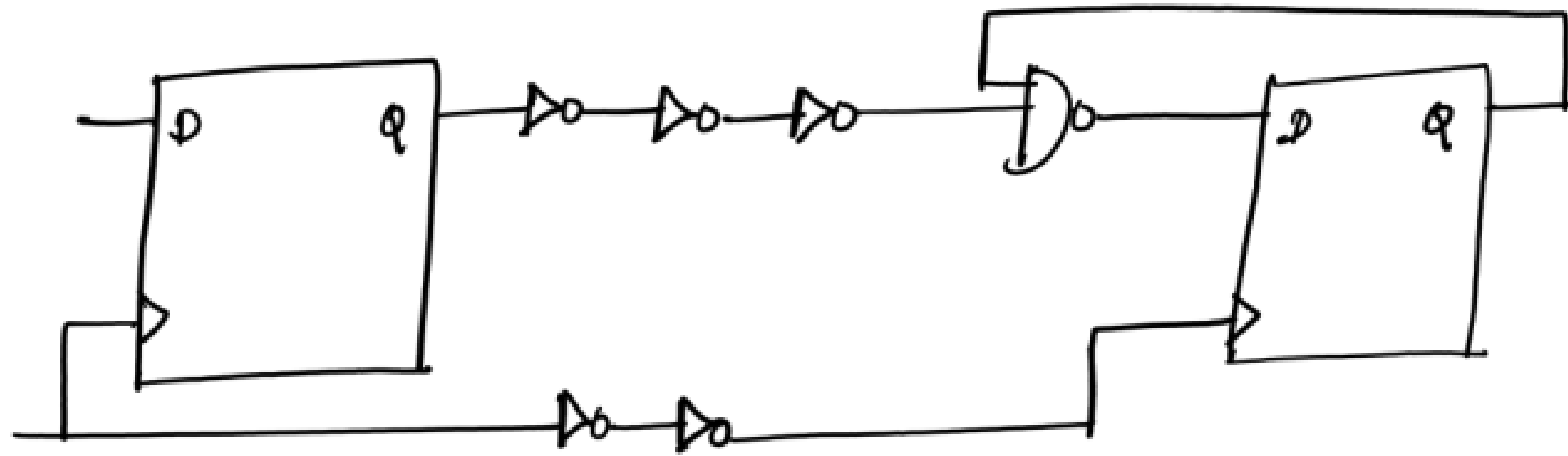
∴ no hold violation

Hold slack :- AT - RT

$$= 4 - 3 = +1 \text{ ns}$$

∴ Hold slack is positive, this path does not have
hold violation

Q : Calculate the max frequency of operation



$$T_{cq} = 2ns$$

$$T_{inv} = 1ns$$

$$T_{setup} = 3ns$$

$$T_{rand} = 2ns$$

$$T_{hold} = 6ns$$

Ans :- $T_{combo} = 1 + 1 + 1 + 2 = 5ns$

$$\Delta_1 + T_{cq} + T_{combo} \leq T_{clk} - T_{setup} + \Delta_2$$

$$0 + 2 + 5 \leq T_{clk} - 3 + 1 + 1$$

$$7 \leq T_{clk} - 1$$

$$7 + 1 \leq T_{clk}$$

$$8ns \leq T_{clk}$$

$$f_{max} = \frac{1}{8 \times 10^{-9}} Hz$$

$$= 0.125 \times 10^9 Hz$$

$$= 0.125 \times 10^3 \times 10^6 Hz$$

$$f_{max} = 125 MHz$$